

Chapter 4

Introduction to Quantitative Forecasting

Currently, both public and private sector organizations place significant emphasis on organizational data. This involves collecting historical numerical data related to operational performance, spanning from the past to the present, which is often disclosed to the public. Such data is typically time-dependent (time series), encompassing examples like daily closing stock prices, monthly tourist arrivals in Thailand, monthly values of agricultural exports, quarterly department store sales, and annual graduation statistics. By employing accurate and appropriate statistical methods to analyze this data and forecast future values, organizations can utilize the resulting projections to support operational planning and strategic decision-making.

Scope of the Study

- 4.1 Types of Statistical Forecasting Techniques
- 4.2 Types of Data
- 4.3 Factors to Consider in Selecting Forecasting Techniques
- 4.4 Measures of Forecasting Accuracy
- 4.5 Statistical Software Packages for Forecasting
- 4.6 Introduction to Time Series
- 4.7 Time Series Patterns
- 4.8 Steps in Time Series Analysis
- 4.9 Trend Estimation

4.1. Types of Statistical Forecasting Techniques

4.1.1 Qualitative Forecasting Techniques

This forecasting method relies on the opinions of experts or experienced individuals. It is not considered a scientific or statistical approach, as it depends solely on personal judgment.

4.1.2 Quantitative Forecasting Techniques

This forecasting method utilizes data collected from the past to the present for analysis to identify trends or study the relationships

between related variables. Quantitative forecasting is divided into two methods as follows:

4.1.2.1 Time Series Analysis

This method is based on the concept that the past behavior of the variable being forecast should be sufficient to predict its future behavior. Therefore, this forecasting approach relies solely on the historical data of the variable itself to determine future trends, without incorporating other external factors (variables).

4.1.2.2 Regression Analysis

This method is based on the concept that the behavior of the variable to be forecast is determined by external factors (other variables). These factors influence the variable of interest through specific forms of relationships.

4.2 **Types of Data**

4.2.1 **Cross-Sectional Data**

Refers to data collected at a single point in time, such as the sales of 10 companies in 2013.

Company	Sales (Y: million baht)	Advertising Budget (X: million baht)
A	100	5.8
B	80	2.3
C	70	2.1
D	120	4.5
E	50	1.8
F	75	2.7
G	98	4.2
H	132	6.0
I	45	2.0
J	62	1.8

Utilize this data to forecast the company's sales using **regression analysis**.

4.2.2 Time Series Data

Refers to data collected over continuous time periods, such as the annual sales of Companies A, B, and C from 2008 to 2013.

4.2.2.1) Time Series Analysis: The Case of a Single Forecast Variable (Y)

Year	Sales (Y: million baht)		
	A	B	C
2008	85	60	50
2009	90	67	54
2010	93	72	58
2011	98	74	60
2012	95	78	65
2013	100	80	70

Use this data to forecast the sales of **each company** (Company A, B, or C) using **Time Series Analysis**.

4.2.2.2) Regression Analysis for Time Series Data: The case of a forecast variable (Y) and other related factors (X).

Year	Company A	
	Sales (Y: million baht)	Advertising Budget (X: million baht)
2008	85	1.3
2009	90	1.1
2010	93	1.8
2011	98	1.2
2012	95	1.5
2013	100	2.2

Use this data to forecast the sales of Company A using **Regression Analysis for Time Series**.

4.3 Factors to Consider in Selecting Forecasting Techniques

4.3.1 Data Pattern

4.3.2 Time Frame

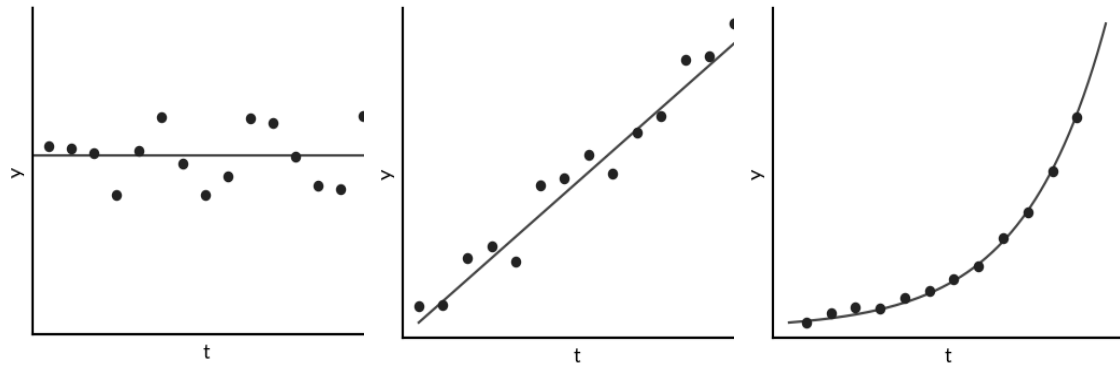
4.3.3 Cost

4.3.4 Accuracy

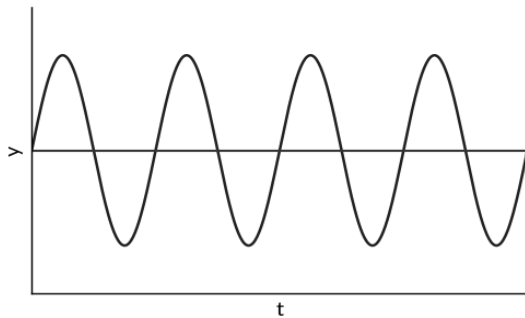
4.3.1 Data Pattern

Refers to historical data patterns, where the analyst must study the characteristics and movement patterns of the time series to determine

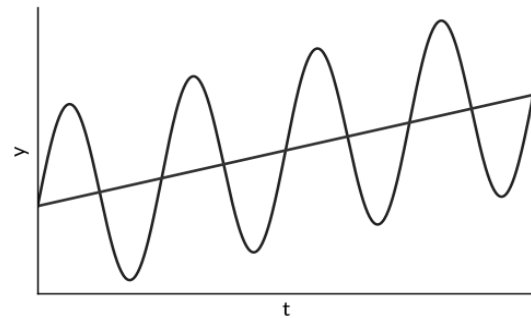
- 1) **Data Trend** or **No Trend** In cases where the data exhibits a trend, it may be **linear** or **curvilinear** (such as quadratic or exponential).



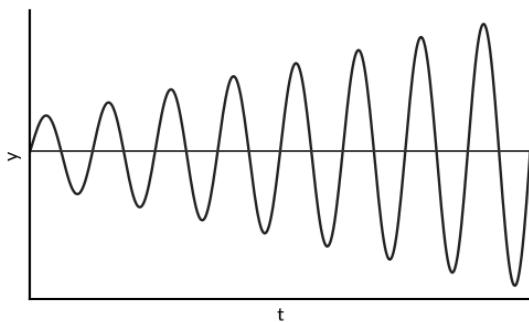
2) Whether the data under consideration exhibits **seasonality** or **no seasonality**.



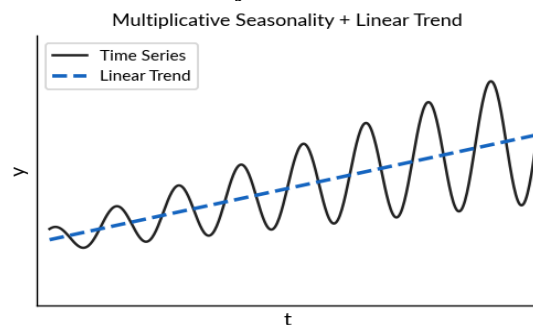
a) Time series with no trend and additive seasonality.



b) Time series with a linear trend and additive seasonality.



c) Time series with no trend and multiplicative seasonality.



d) Time series with a linear trend and multiplicative seasonality.

● **Guidelines for Selecting Time Series Forecasting Methods**

When selecting a time series forecasting method, the analyst should initially examine the graphical characteristics of the time series to determine whether seasonal variation is present. Alternatively, the '**Seasonal Index**' may be considered. If the seasonal index for any season deviates significantly from 100%, it indicates the presence of seasonal variation in the time series. Conversely, if the seasonal indices for all seasons are approximately equal to 100%, it implies that the time series exhibits no seasonal

variation. Furthermore, the movement patterns of the time series must be analyzed to determine the existence of a trend. If a trend is present, its nature must be identified, whether it is linear or curvilinear (e.g., quadratic or exponential). The guidelines for selecting time series forecasting methods can be summarized as follows:

Figure 4.1: Guidelines for Selecting Time Series Forecasting Methods

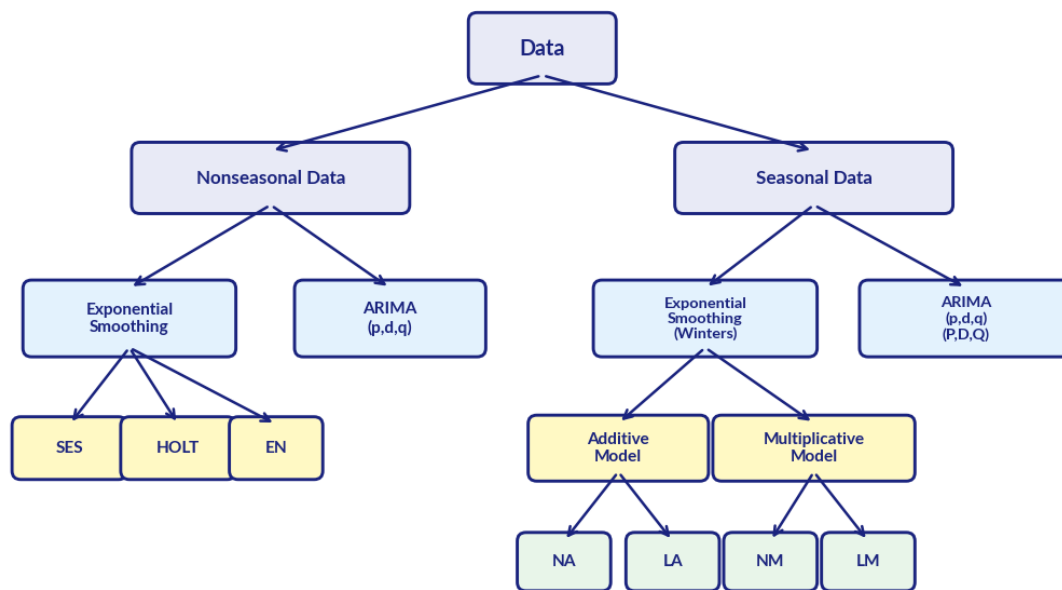


Figure 4.1: Guidelines for Selecting Time Series Forecasting Methods

SES = Simple Exponential Smoothing

The SES method is used to forecast time series data that exhibits **No Trend and No Seasonal Variation**.

The Holt method is used to forecast time series data that exhibits Linear Trend and No Seasonal Variation

The EN method is used to forecast time series data that exhibits Exponential Trend and No Seasonal Variation

The NA method is used to forecast time series data that exhibits No Trend and Additive Seasonal Variation

The LA method is used to forecast time series data that exhibits Linear Trend and Additive Seasonal Variation

The NM method is used to forecast time series data that exhibits No Trend and Multiplicative Seasonal Variation

The LM method is used to forecast time series data that exhibits Linear Trend and Multiplicative Seasonal Variation

4.3.2 Time Frame

- 1) **Data Collection Interval for Analysis** The data collection period may be measured in units of **days, weeks, months, quarters, or years.**
- 2) **Forecasting Horizon** The time period for future forecasting can be categorized into three durations as follows:
 - **Short-Term**
Refers to a forecast covering **1 to 3 future periods** (e.g., days, weeks, months, or quarters).
 - **Medium-Term**
Refers to a period ranging from **3 months to 2 years**. A common example of medium-term forecasting is production planning for a company.
 - **Long-Term**
Refers to a period of **2 years or more**. This is typically used for high-level strategic planning, such as national economic and social development planning.

4.3.3 Cost

Forecasting involves costs ranging from **data collection** to the computational expenses associated with applying the most appropriate forecasting technique for the available data.

4.3.4 Accuracy

A single time series dataset can be analyzed using various techniques. The optimal forecasting method is the one that yields the highest accuracy. This is determined by measuring the error between **the actual value** (Y_i) and **the forecasted value** ($\hat{Y}_i$). There are several metrics commonly used to measure this error, such as:

- 1) Mean Absolute Deviation: MAD
- 2) Mean Squared Error: MSE
- 3) Mean Absolute Percentage Error: MAPE

4) Mean Percentage Error: MPE

4.4. Measuring Forecast Accuracy

Multiple forecasting methods can be applied to a single dataset. The selection of the best and most appropriate method is determined by the forecast error. The objective is to find a method where the forecast error is minimized while ensuring the forecasting model exhibits minimal complexity (Principle of Parsimony). Forecast error can be measured using the following four methods:

4.4.1 Mean Absolute Deviation: MAD

$$MAD = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

4.4.2 Mean Squared Error: MSE

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} \quad (\text{has units of } y)^2$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

Root Mean Squared Error (RMSE): The square root of the Mean Squared Error, which has the same unit of measurement as y

4.4.3 Mean Absolute Percentage Error: MAPE

This value measures forecast accuracy by comparing the size of the error to the actual value. It allows for comparing the accuracy of different time series that have different units, using the same forecasting method. The formula is:

$$MAPE = \frac{\sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|}{n} \times 100$$

MAPE is dimensionless (unit-free) and is typically expressed as a percentage.

4.4.4 Mean Percentage Error: MPE

$$MPE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)/y_i}{n} \times 100$$

If $MPE \rightarrow 0$ the forecasting method yields an

unbiased forecast $\Rightarrow \hat{y}_i \cong y_i$

If $MPE \rightarrow -\infty$ the forecasting method yields an $\Rightarrow \hat{y}_i > y_i$

If $MPE \rightarrow +\infty$ the forecasting method yields an $\Rightarrow \hat{y}_i < y_i$

where y_t is the **actual value** of the time series at time t

$\hat{y}_t$ is the **forecasted value** of the time series at time t

and e_t is the **forecast error** at time e_t , defined as $e_t = y_t - \hat{y}_t$

4.5. Statistical Software Packages for Forecasting

Forecasting involves costs ranging from data collection to the computation or data analysis required by the most appropriate forecasting technique for the available data. Consequently, statistical software packages are crucial and essential for quantitative data analysis, as many forecasting techniques involve extensive, complex, and multi-step computational processes. Learners should gain an understanding of various forecasting techniques and study how to utilize statistical software packages that incorporate these techniques for processing data and interpreting the results. There are numerous statistical software packages available, such as:

- 1) SPSS (Statistical Package for Social Sciences)
- 2) SAS (Statistical Analysis System)
- 3) Minitab
- 4) TSP
- 5) EView
- 6) Forecast Plus

Year	Sales (y)	Sales forecast value ($\hat{y}$)		
		Method 1	Method 2	Method 3
2543	90	97	79	91
2544	102	108	101	106
2545	115	109	117	129
2546	120	112	100	117
2547	129	136	118	130
2548	142	148	141	146
2549	156	150	158	170
2550	183	175	163	180

Example A company collected annual sales data, in millions of Baht, from the year 2000 to 2007 (B.E. 2543–2550) to forecast future sales. The analyst employed three different forecasting techniques, yielding the forecasted values ($\hat{y}$) shown in the table. Calculate the accuracy metrics to compare and determine which forecasting technique is the most appropriate.

Consider the forecast using Method 1

$$1. \quad MAD = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

$$= \frac{|90 - 97| + |102 - 108| + |115 - 109| + |120 - 112| + |129 - 136| + |142 - 148| + |156 - 150| + |183 - 175|}{8}$$

$$= 6.75 \text{ million baht}$$

$$2. \quad MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

$$= \frac{(90 - 97)^2 + (102 - 108)^2 + (115 - 109)^2 + (120 - 112)^2 + (129 - 136)^2 + (142 - 148)^2 + (156 - 150)^2 + (183 - 175)^2}{8}$$

$$= \frac{(90 - 97)^2 + (102 - 108)^2 + (115 - 109)^2 + (120 - 112)^2 + (129 - 136)^2 + (142 - 148)^2 + (156 - 150)^2 + (183 - 175)^2}{8}$$

$$(\text{million baht})^2$$

$$\sqrt{MSE} = \sqrt{46.25} = 6.80 \text{ million baht}$$

$$3. \text{MAPE} = \frac{\sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|}{n} \times 100$$

$$= \frac{\left| \frac{90-97}{90} \right| + \left| \frac{102-108}{102} \right| + \left| \frac{115-109}{115} \right| + \left| \frac{120-112}{120} \right| + \left| \frac{129-136}{129} \right| + \left| \frac{142-148}{142} \right| + \left| \frac{156-150}{156} \right| + \left| \frac{183-175}{183} \right|}{8} \times 100$$

$$4. \text{MPE} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)/y_i}{n} \times 100$$

$$= \frac{\frac{(90-97)}{90} + \frac{(102-108)}{102} + \frac{(115-109)}{115} + \frac{(120-112)}{120} + \frac{(129-136)}{129} + \frac{(142-148)}{142} + \frac{(156-150)}{156} + \frac{(183-175)}{183}}{8} \times 100$$

$$= \frac{-0.03210}{8} \times 100 = -0.004013 \times 100 = -0.4013\%$$

Comparison of MAD, MSE, MAPE, and MPE for Three Forecasting Methods

Accuracy values	Forecasting Technique by (Unit: Million Baht)		
	Method 1	Method 2	Method 3
1. <i>MAD</i>	6.75 Unit	8.50 Unit	5.50 Unit *
2. <i>MSE</i>	46.25 Unit ² *	131.50 Unit ²	55.50 Unit ²
$RMSE = \sqrt{MSE}$	6.80 Unit*	11.47 Unit	7.45 Unit
3. <i>MAPE</i>	5.4267%	6.6313%	4.2390%*
4. <i>MPE</i>	-0.4013%*	5.8761%	-3.2042%

Based on the annual sales data, it is evident that the value of *MSE*, *MSE* (or Root Mean Square Error: RMSE) and $\sqrt{MSE}$ which is a metric used to measure forecast error, is the lowest for **Method 1**, and is lower than those of Method 2 and Method 3. However, when examining the forecast using **Method 3**, it is observed that the error measurements *MSE* and *MPE* are the lowest, being lower than those of Method 1 and

Method 2. Therefore, the analyst may choose either **Method 1** or **Method 3**, depending on which error measurement criterion is prioritized.

4.6 Introduction to Time Series

4.6.1 Definition of "Time Series"

"Time Series" This refers to a collection of data gathered in chronological order (or sequentially) over a specific period of time. Examples include:

- Monthly sales of a department store
- Quarterly number of foreign investors in the Stock Exchange of Thailand (SET)
- Annual cigarette production volume of the Tobacco Authority
- Monthly volume of agricultural exports
- Annual number of new entrants into the labor market
- Annual population of Thailand
- Daily stock prices of individual companies in the stock exchange

4.6.2 Objectives of Time Series Analysis

1. To plot a time series graph:

The Horizontal Axis x represents consecutive time periods, such as days, weeks, months, quarters, or years.

The Vertical Axis y represents the values of the variable under consideration.

2. To identify components: Based on the graph constructed in item 1, determine which of the **4 components** are present in

the data y

- 1) Trend : T
- 2) Seasonal Variation : S
- 3) Cyclical Variation : C
- 4) Irregular Variation : I

3. To calculate and decompose: To calculate and isolate the values of each component T, S, C and I from the dataset y .

4.6.3 Benefits of Time Series Analysis

1. To examine the influences or various components inherent in a time series dataset.
2. To forecast future values of the variable y .
3. To support future planning and decision-making processes for various organizations.

4.6.4 Components of a Time Series

Generally, time series data in business and economics are composed of **four key components**, namely:

1. Long-Term Trend : T
2. Seasonal Variation : s
3. Cyclical Variation : c
4. Irregular Variation : I

4.6.4.1 Long-Term Trend : T

This represents the movement of a time series over a relatively long period of time. Trends are observed in almost all time series data.

- Changes in population
- Gross National Product (GNP)
- Export value of major agricultural products

The study of **Trend** involves analyzing the changes in a time series from the past up to the most recently collected data. It aims to determine whether the direction is **increasing (upward)** or **decreasing (downward)**, and whether the pattern is **linear** or **curvilinear**.

Monthly = Occurring every month

Annual = Occurring every year - *Formal*

Yearly = Occurring every year

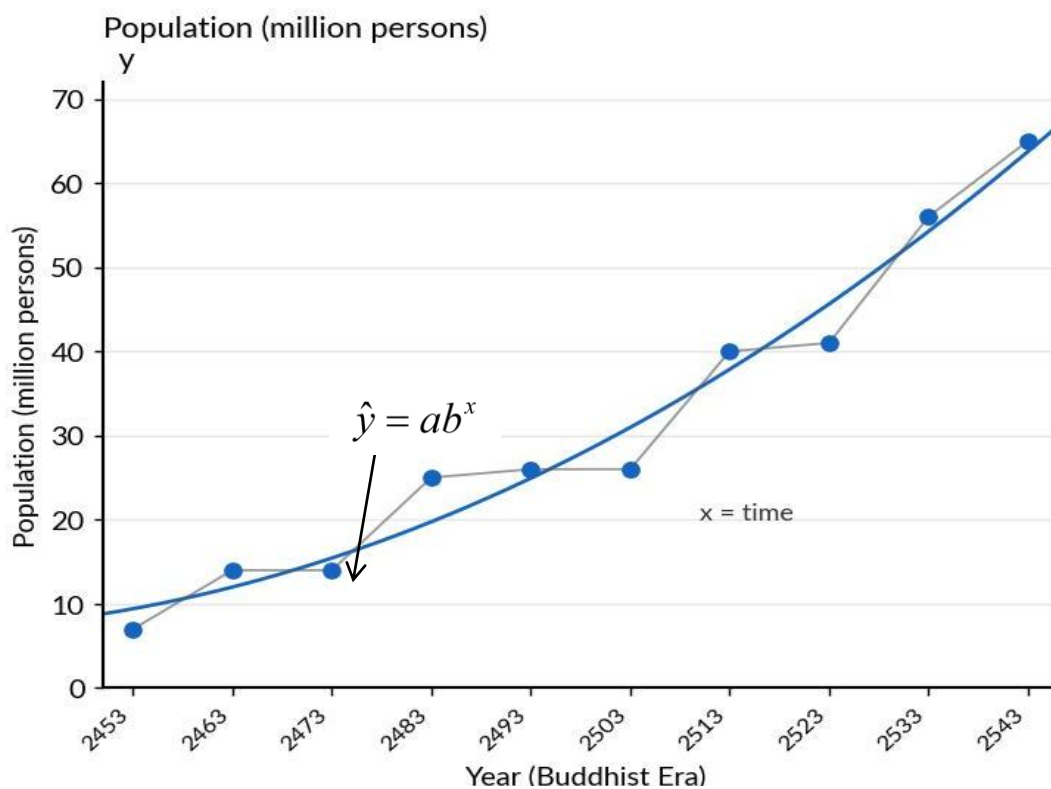
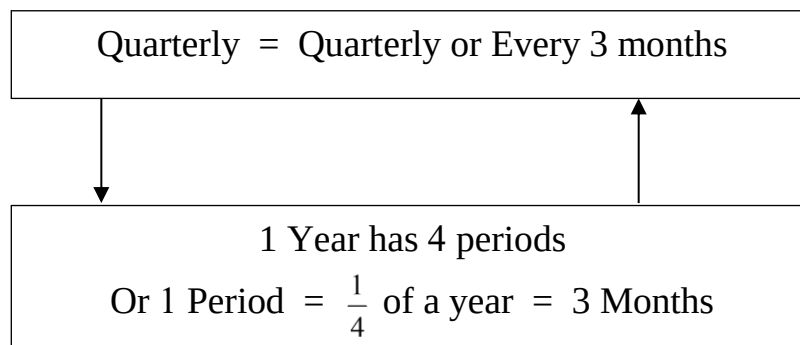


Figure 1: Shows the increasing trend of the Thai population from 1910 to 2000 (B.E. 2453–2543) (Linear Trend).

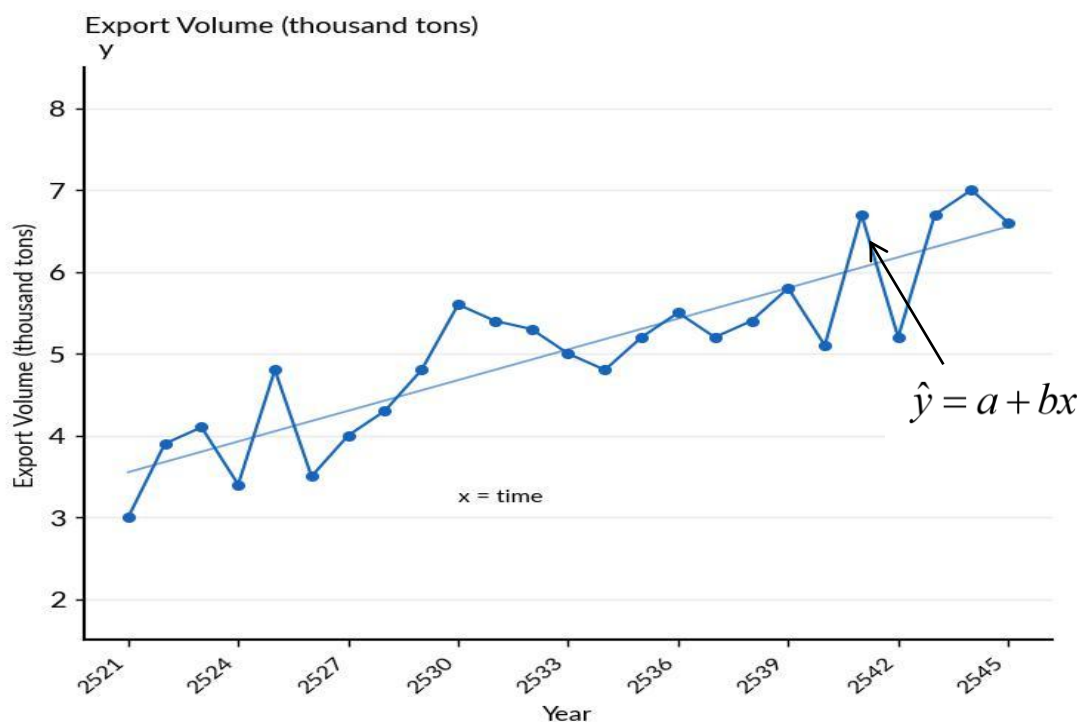


Figure 2: Shows the trend of Thailand's rice export volume from 1978 to 2002 (B.E. 2521–2545) (Linear Trend).

4.6.4.2 Seasonal Variation : s

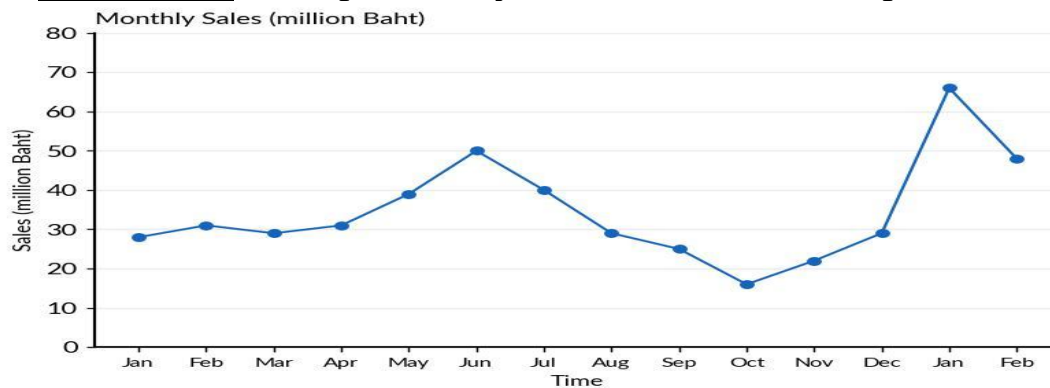
This refers to repetitive fluctuations (up and down movements) that occur over a fixed period of time. It is typically observed in time series data collected over intervals of **less than one year**. This may manifest as daily variations repeating every week, or monthly variations repeating every year.

- Department store sales volume is typically higher on Saturdays compared to other days.
- Sales volume of winter coats during the winter season.
- Tap water consumption during the summer.
- Umbrella sales during the rainy season.

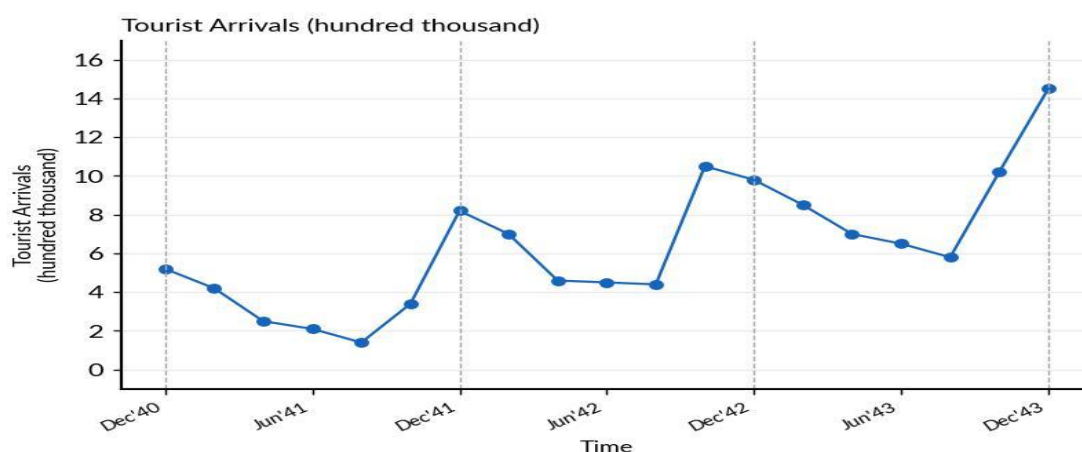
Seasonal influences cause data to deviate from the general trend or normal pattern. This variation is commonly measured using an index known as the '**Seasonal Index**'.

Time series data that may exhibit seasonality (Seasonal : s)

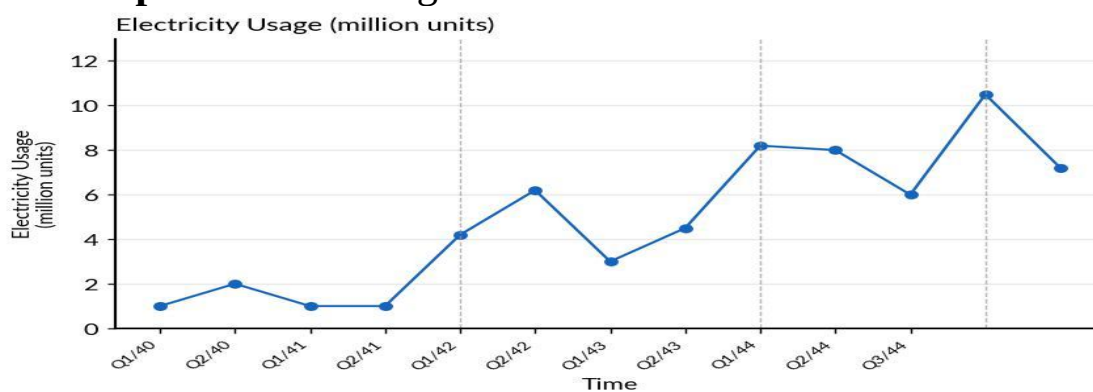
1. Daily Data Example: Daily sales volume of a department store



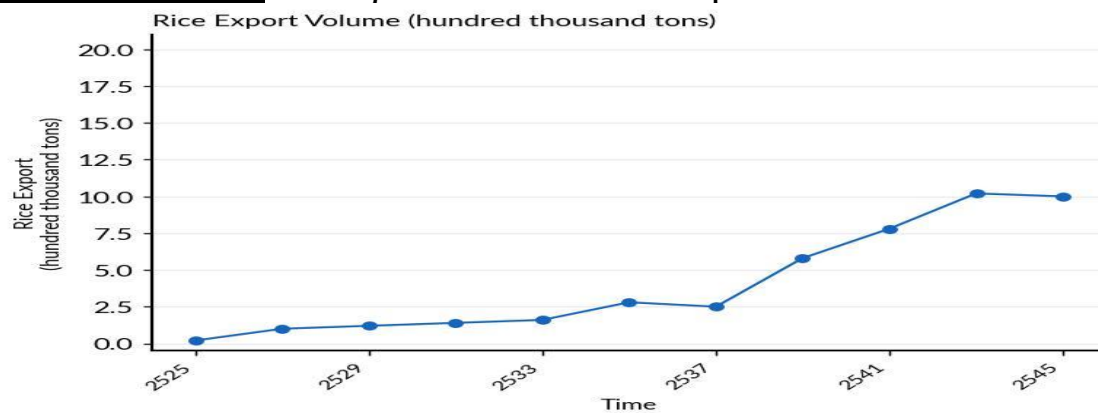
2. Monthly Data: Example: Monthly number of tourists visiting Thailand



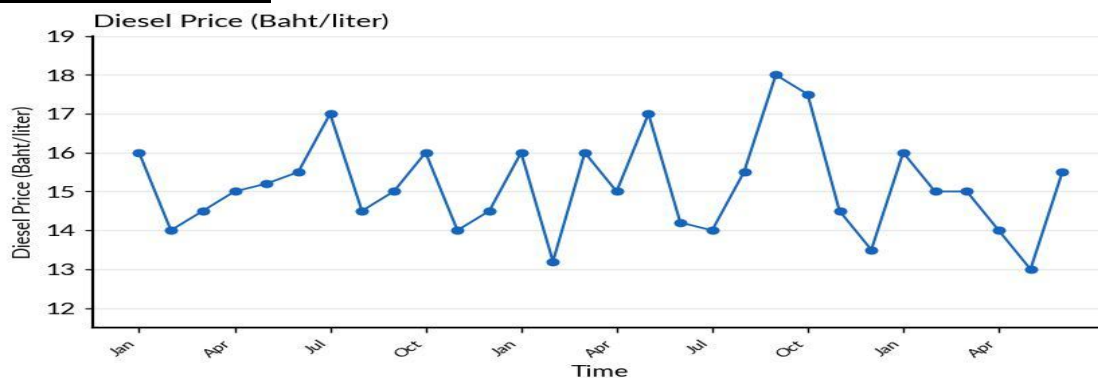
3. "Quarterly" Data: Example: Quarterly household electricity consumption in the Bangkok area.



4. "Annual" or "Yearly" Data: These data do not exhibit Seasonality (S) Example: Annual rice export volume.

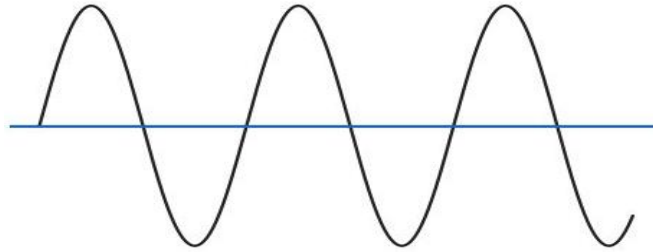


5. Daily Diesel Price Data (Baht/Liter) from a specific gas station. From the graph, it is observed that there is no Seasonal Variation (S) component.

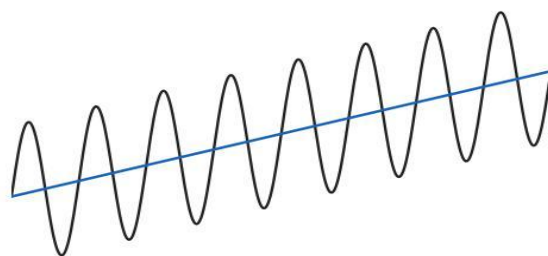


1) Additive Seasonal Variation

This refers to a time series where the magnitude of the seasonal fluctuations remains constant over time.

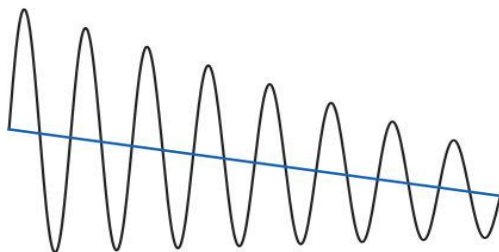


Additive S + No Trend



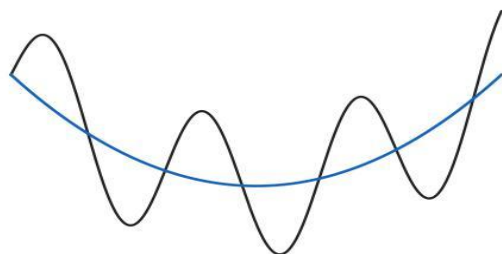
Additive S + Linear Trend

Additive Seasonality Increasing Linear Trend



Additive S + Linear Trend

Additive Seasonality Decreasing Linear Trend



Additive S + Curvilinear Trend

Additive Seasonality Changing Trend

Figure 3: Illustrates the Additive Seasonal Pattern.

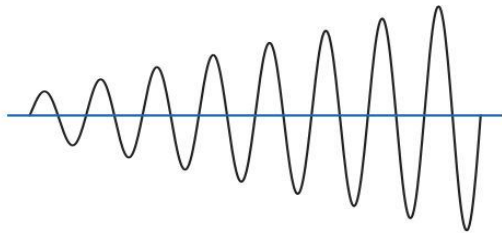
* **Note:** If the Seasonality (S) is additive, the model employed is

$$y = T + S + C + I$$

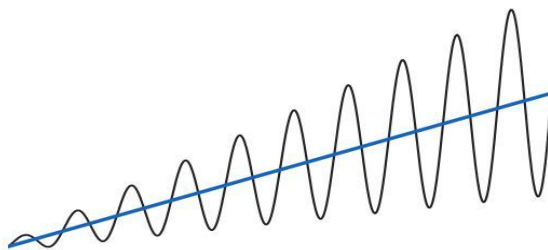
(Additive Model).

2) Multiplicative Seasonal Variation

This refers to a time series where the **magnitude of the seasonal fluctuations changes** over time.

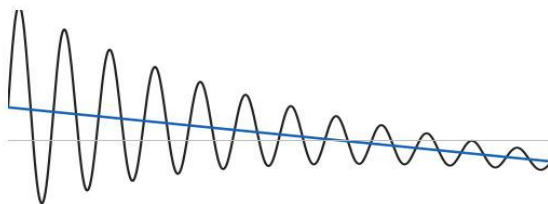


Multiplicative $S \times$ No Trend



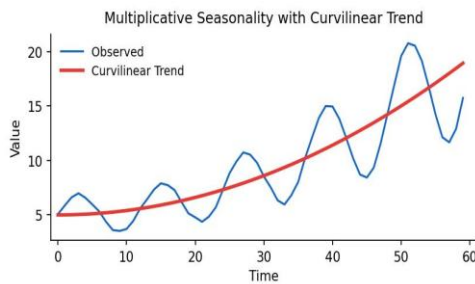
Multiplicative $S \times$ with Linear
Trend

Multiplicative Seasonality Increasing Linear Trend



Multiplicative $S \times$ with Linear
Trend

Multiplicative Seasonality Decreasing Linear Trend



Multiplicative $S \times$ with Curvilinear Trend

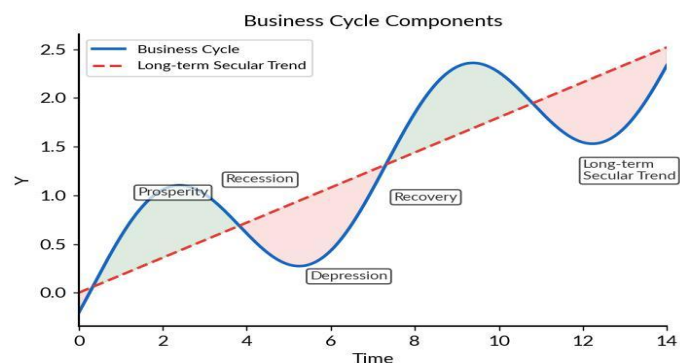
Multiplicative Seasonality Changing Trend

Figure 4: Illustrates the Multiplicative Seasonal Pattern.

* **Note:** If the Seasonality (S) is Multiplicative, the model employed is $y = T \times S \times C \times I$ (Multiplicative Model)

4.6.4.3 Cyclical Variation : C

This refers to **wavelike fluctuations** (up and down movements) in a time series that occur over a period **longer than one year**. Typically, cyclical variation is driven by the fluctuations of the **Business Cycle** (or Economic Cycle), which consists of four stages:



- **Prosperity** (or Economic Boom)
- Recession
- Depression
- Recovery

These are consecutive economic phases where the **duration of each phase in the cycle is unpredictable**

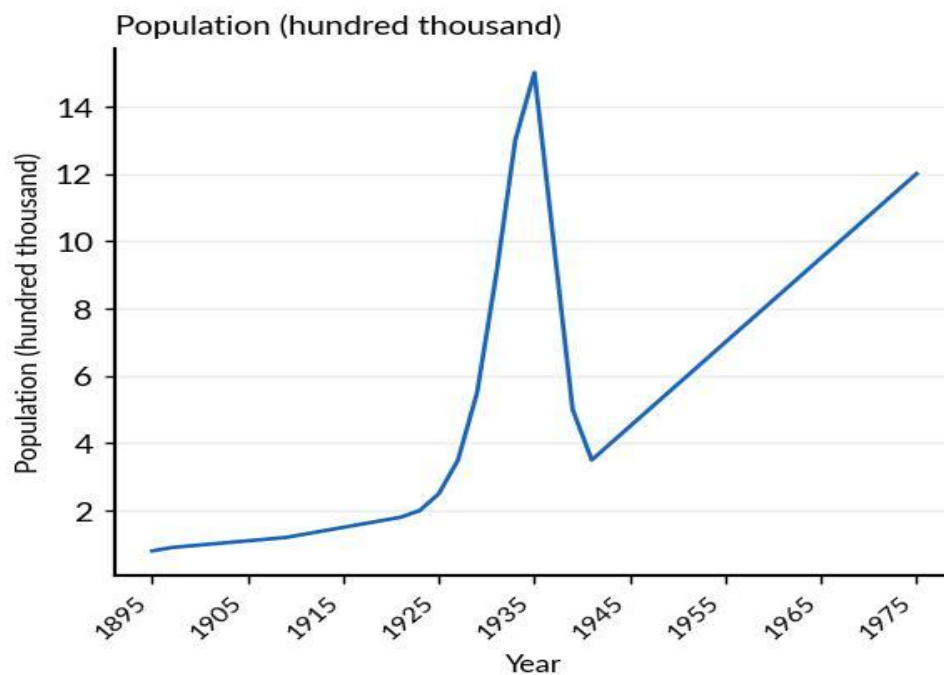
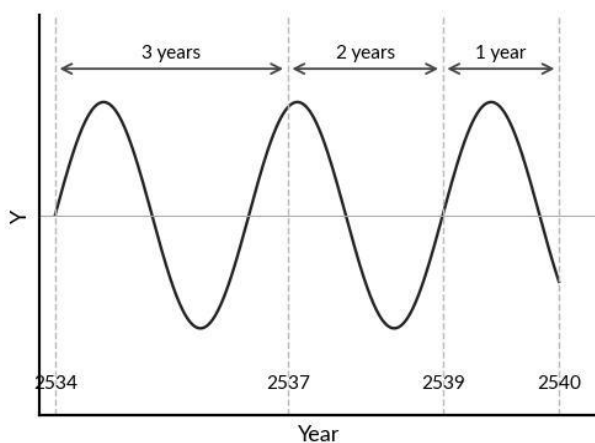


Figure 5: Number of Unemployed Persons in the United States, 1900–1976

Key Differences between Cyclical Variation and Seasonal Variation



Cyclical Variation Cannot be predicted (unpredictable). This is because the duration of each cycle is **irregular** (unequal), and the magnitude of the peaks and troughs varies significantly.

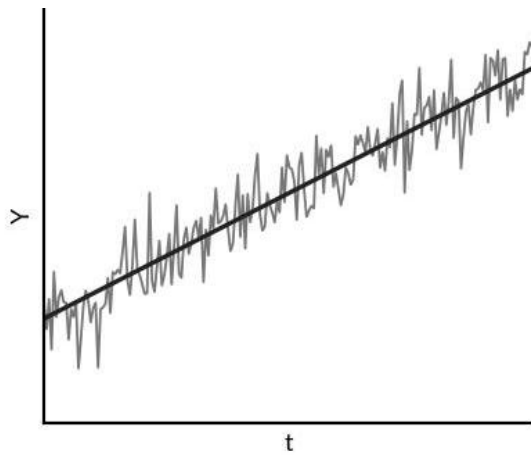
Seasonal Variation Can be predicted (forecasted). This is because the duration of the repetitive patterns (cycles) is **constant** (fixed length).

4.6.4.4 Irregular Variation : *I*

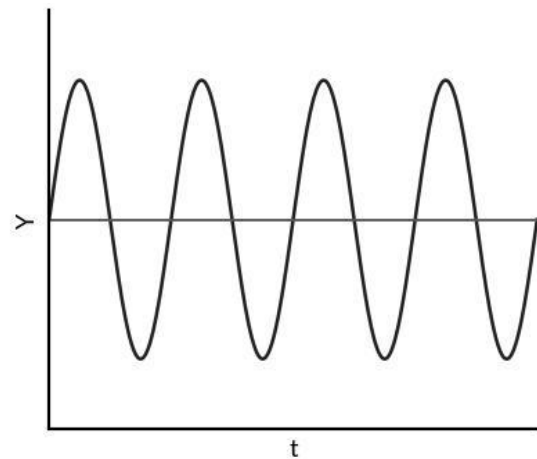
This refers to **unpredictable** (or erratic) up-and-down movements occurring over **short periods of time**. Causes may include:

- Abnormal weather conditions.
- Flash floods.
- Labor strikes.
- Wars.
- Sudden changes in fashion.

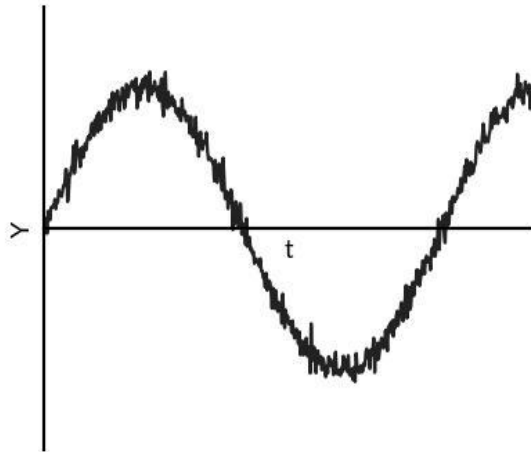
Irregular variation is characterized by **non-systematic movements** (randomness). Consequently, phenomena arising from these abnormal conditions **cannot be predicted** in advance.



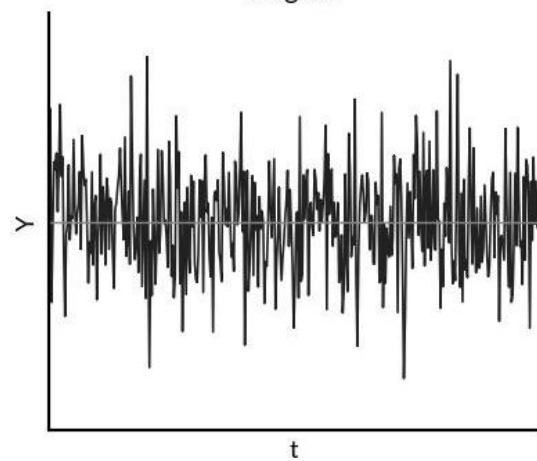
a) Trend



b) Seasonal



c) Cyclical



d) Irregular

Figures (a)–(d) illustrate the components of a time series.

Graphs Illustrating Time Series Components

Figure 6: This graph shows the **Linear Trend** of sales for a department store, 1988-1996.

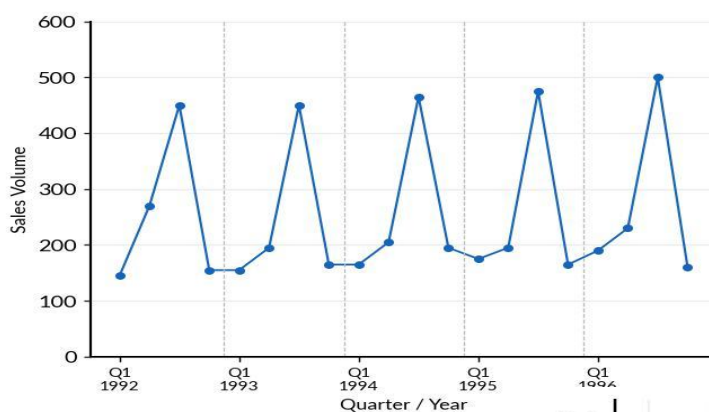
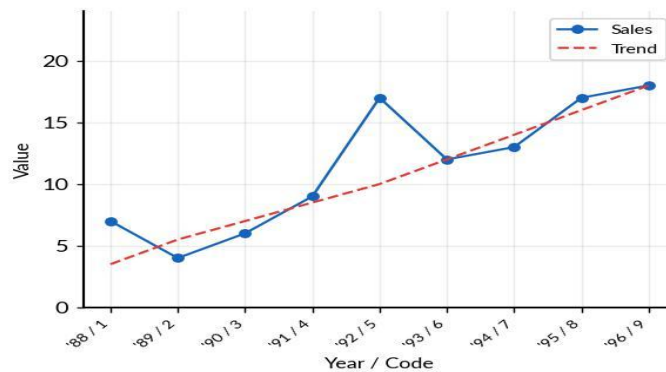


Figure 7: This graph shows the **Seasonal Variation** (Quarterly) of the number of unemployed persons, 1992-1996.

Figure 8: This graph shows the **Cyclical Variation** of new housing starts (or new houses built) in a specific city, 1959-1991.

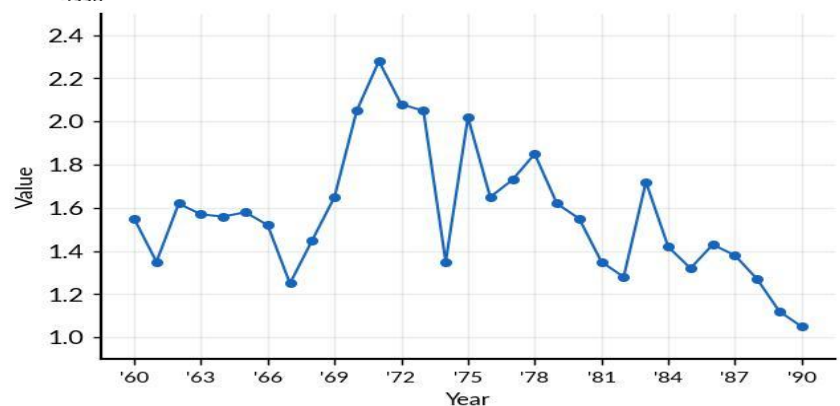
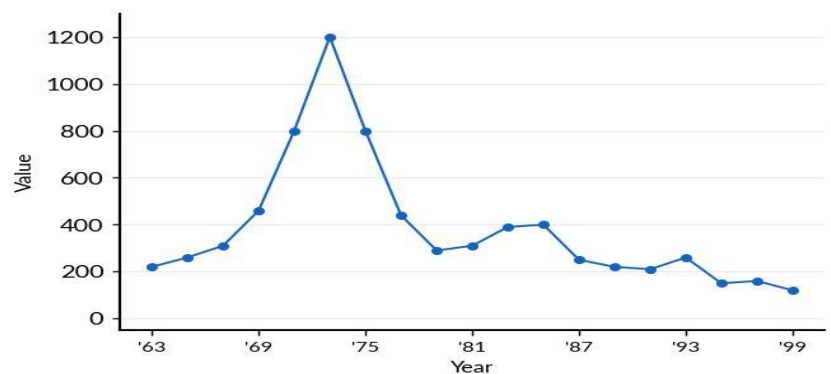


Figure 9: This graph shows the **Irregular Variation** of the number of imported handguns, 1963-1992.



4.7 Time Series Model

1) Additive Model

$$Y = T + S + C + I$$

The Additive Model assumes that the four components are **independent of each other**. This implies that a change in any single component, whether an increase or a decrease, does not affect the remaining components. For instance, the Trend has no influence on Seasonal Variation or Cyclical Variation. Furthermore, the unit of each component (T , S , C and I) is the same as the unit of the observed value (Y).

2) Multiplicative Model

$$Y = T \times S \times C \times I$$

The Multiplicative Model assumes that the **four components are interdependent** (dependent on each other). That is, they are related and not independent. This implies that if the value of any single component changes, whether increasing or decreasing, it will affect the remaining components. This model is commonly used with economic and business time series data, because a change in one business or economic factor typically impacts other factors; such as: a change in bank loan interest rates (whether an increase or decrease) will significantly impact numerous other business factors. Regarding units: The unit of the Trend (T) component is the same as the unit of the observed data (Y), whereas the components S , C and I are expressed as percentages (or indices).

4.8 Time Series Analysis Procedure Using the method known as "**Time Series Decomposition**," which is classified as a **Classical Method**.

In this context, we consider the **Multiplicative Model** :

$$Y = T \times S \times C \times I$$

1) For Daily, Monthly, or Quarterly Data: First, estimate the Seasonal Variation (s) Then, deseasonalize the data (remove s) by dividing the original time series (Y) by the seasonal component (s). The remaining components will be $T \times C \times I$. In the case of Annual Data, there is no Seasonal component (s)

2) Take the result from Step 1 ($T \times C \times I$) and apply a **moving average** (specifically, a 3-period moving average applied twice) to eliminate the influence of **Irregular Variation** (I). This yields the **Trend-Cycle component** ($T \times C$). To isolate the irregular component, divide the result from Step 1 ($T \times C \times I$) by the smoothed data ($T \times C$). The remaining component is I .

3) Take the Trend-Cycle ($T \times C$) obtained in Step 2 and estimate the Trend (T) using the Method of Least Squares. Then, remove the Trend (T) by dividing the Trend-Cycle ($T \times C$) by the estimated Trend (T). The remaining component is the Cyclical Variation (C).

4) Forecast the value of Y by multiplying the estimates of all four components ($\hat{Y} = T \times S \times C \times I$). (When forecasting future values of Y , if the $C \times I$ is unknown, it is typically estimated as 1.)

4.9 Trend Estimation (T) There are two methods:

- 1) Moving Average Method
- 2) Method of Least Squares

4.9.1 Method of Moving Average : MA

This is a data smoothing technique. The result of the moving average yields an estimate of the Trend (T).

The Moving Average Method involves the following steps:

- 1) Divide the data into groups based on the desired moving-average period. For example:
 - Groups of **3 months** (if the data is monthly)
 - Groups of **5 years** (if the data is annual), etc.
- 2) Calculate the average for each group. Specifically, once the average for the first group has been determined, remove the first data point of that group and replace it with the next data point (the one immediately following the last member of the first group); then, calculate the new average to obtain the value for the second group. Repeat this process continuously until the last data point is reached.

Centered Moving Average

Sometimes, when using the moving average method to estimate the Trend, if the moving average period is an even number, such as 4 periods, 6 periods, or 12 periods:

In such cases, the calculated moving average will not fall exactly in the middle of a specific period or year (it falls between two time points). Therefore, a secondary moving average of 2 periods must be performed to align the data. This process is called a '**Centered Moving Average**'.

- Note:**
1. If the data is Annual (Yearly) and a 5-year Moving Average is performed: The result of the moving average will yield data containing the Trend estimate (T). (The influence of C and I will be eliminated.)
 2. If the data is Monthly and a 12-month Moving Average is performed: The result of the moving average will yield data containing the components $T \times C$ (Trend and Cyclical). (The influence of S and I will be eliminated.)
 3. If the data is Quarterly (1 year has 4 quarters) and a 4-quarter Moving Average is performed: The result of the moving average will yield data containing the components $T \times C$ (Trend and Cyclical). (The influence of S and I will be eliminated.)

Trend Estimation (T) using a 3-Point Moving Average Method

Example: Given that following data represents a time series over 12 consecutive periods, where the data Y consists of values Y and $a, b, \dots, k$ respectively,

Estimate **the Trend (T)** using the **3-Point Moving Average Method** (using a moving average period = 3).

Time	Data Y	Trend Value (T) using 3-Point Moving Average
1	a	-
2	b	$\frac{a+b+c}{3} = \text{-----}$
3	c	$\frac{b+c+d}{3} = \text{-----}$
4	d	$\frac{c+d+e}{3} = \text{-----}$
5	e	$\frac{d+e+f}{3} = \text{-----}$
6	f	$\frac{e+f+g}{3} = \text{-----}$
7	g	$\frac{f+g+h}{3} = \text{-----}$
8	h	$\frac{g+h+i}{3} = \text{-----}$

Ti me	Data Y	Trend Value (T) using 3-Point Moving Average
9	i	$\frac{h+i+j}{3} = \text{-----}$
10	j	$\frac{i+j+k}{3} = \text{-----}$
11	k	$\frac{j+k+l}{3} = \text{-----}$
12	l	-

Disadvantage : It can be seen that the **Trend values (T)** estimated by the **Moving Average (MA)** method will be **missing** for the data in the **beginning** and the **end** sections.

Example: The following data represents the annual profit (in units of hundred thousand Baht) of a department store from B.E. 2529–2540. Estimate the trend using a **5-Year Moving Average** method.

Year	Profit (100,000 Baht)	5-Year Moving Total	<u>5-Year Moving Average</u>
2529	2	-	-
2530	4	-	-
2531	5	$2+4+5+7+8=26$	$\frac{26}{5} = 5.2$
2532	7	$4+5+7+8+6=30$	$\frac{30}{5} = 6.0$
2533	8	$5+7+8+6+8=34$	$\frac{34}{5} = 6.8$
2534	6	$7+8+6+8+11=40$	$\frac{40}{5} = 8.0$
2535	8	$8+6+8+11+13=46$	$\frac{46}{5} = 9.2$
2536	11	$6+8+11+13+14=52$	$\frac{52}{5} = 10.4$
2537	13	$8+11+13+14+11=57$	$\frac{57}{5} = 11.4$
2538	14	$11+13+14+11+14=63$	$\frac{63}{5} = 12.6$
2539	11	-	-
2540	14	-	-

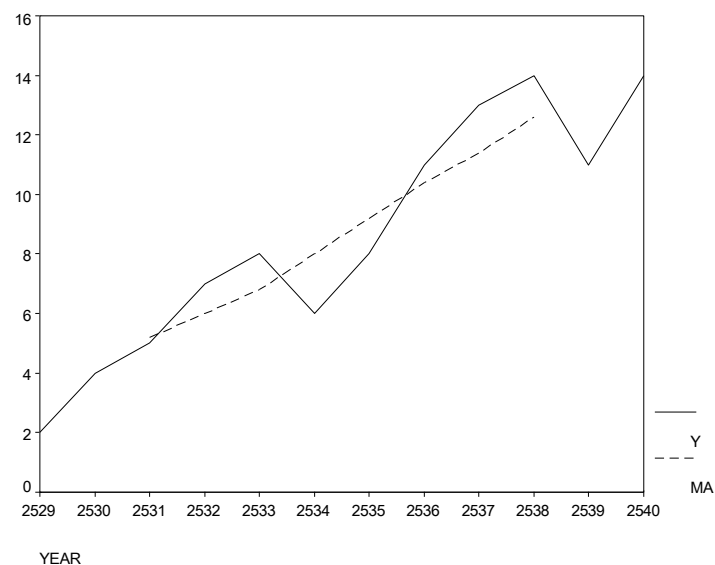


Figure 10 : shows Profit (Y) and Estimated Trend ($\hat{Y}$) of profit using the **5-Year Moving Average Method**.

Example: From the following data, estimate the trend using the **4-Quarter Moving Average Method**.

Year	Quarter	y	<u>Moving Average</u> 4 Quarter (MA)	Centered Moving Average (CMA)
2531	1	2978		
	2	2790	3409.00	3512.625
	3	3251	3616.25	3689.250
	4	4617	3762.25	3881.875
2532	1	3807	4001.50	4352.125
	2		4702.75	5378.750
	3		6054.75	6355.500
	4		6656.25	7329.875
2533	1		8003.50	8638.375
	2		9273.25	
	3			
	4			

Year	Quarter	y	<u>Moving Average</u> 4 Quarter (MA)	Centered Moving Average (CMA)
2533	3	4208		
	4	7422		
	1	9215		
	2	5780		
	3	9837		
	4	12261		

Disadvantages of Trend Estimation using the Moving Average Method

1. **No Trend Equation:** There is no mathematical trend equation (function) available to be used for forecasting future trend values.
2. **Missing Values:** Some trend values at the **beginning** and the **end** of the time series will be missing (lost).

Since the data obtained from the moving average at the **beginning** and the **end** sections will be missing, it is not possible to study the **Trend** (T) of the data in the future. Therefore, we will study the **Trend** (T) by constructing equations: 1) Linear Case : $\hat{y} = a + bx$ 2) Curvilinear Quadratic Case : $\hat{y} = a + b_1x + b_2x^2$ 3) Curvilinear Exponential Case : $\hat{y} = ab^x$ And from these trend forecasting equations, we can study or forecast the **Trend** (T) in the future.

4.9.2 Trend Estimation (Trend) by Method of Least Squares

The **Method of Least Squares** (LS) is a very popular method for finding trend estimation equations. This method finds the trend such that the sum of the squared differences between the collected data (Y) and the estimated trend value ($\hat{Y}$) or $\sum(Y - \hat{Y})^2$, is minimized. We will consider 3 trend models:

1. Linear Model

The forecasting equation is $\hat{Y}_T = a + bx$. This is a linear trend estimation equation (T). By the Method of Least Squares, the values of a and b can be found from the "**Normal Equations**":

$$\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i \quad (1)$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad (2)$$

By solving equations (1) and (2), we obtain a and b as follows:

$$b = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

and
$$a = \bar{y} - b\bar{x}$$

Where:
$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad \text{and} \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

And x = time, $x = 1, 2, 3, \dots, n$

To use the **SPSS program** to find the trend estimation equation (T) for **time series data**, use the command: using the command Analyze → Curve Estimation when defining the variables to be analyzed, which in this case are the **Dependent variable (y)** and the **Independent variable (x)**, which is Time. SPSS will substitute $x = 1, 2, 3, \dots, n$ to calculate the trend estimation equation according to the selected model.

2. Quadratic Model

The forecasting equation is $\hat{Y}_T = a + b_1x + b_2x^2$. This is a Non-Linear trend estimation equation called a **Second-Degree Polynomial**. Using the Method of Least Squares, the values of a , b_1 and b_2 can be found from the '**Normal Equations**' as follows:

$$\sum y = na + b_1 \sum x + b_2 \sum x^2 \quad (1)$$

$$\sum xy = a \sum x + b_1 \sum x^2 + b_2 \sum x^3 \quad (2)$$

$$\sum x^2 y = a \sum x^2 + b_1 \sum x^3 + b_2 \sum x^4 \quad (3)$$

Where: a = estimated Y intercept

b_1 = estimated linear effect on Y

b_2 = estimated curvilinear effect on Y

Solve the normal equations to find a , b_1 and b_2 where x = time and $x = 1, 2, 3, \dots, n$.

3. Exponential Model

The forecasting equation is $\hat{Y}_T = ab^x$. This is a Non-Linear trend estimation equation. It can be transformed into a linear model by taking the natural logarithm ($\ln$) of both sides:

$$\ln y = \ln a + x(\ln b)$$

Find a and b from the '**Normal Equations**':

$$\sum \ln y = n(\ln a) + (\ln b) \sum x \quad (1)$$

$$\sum x \ln y = (\ln a) \sum x + (\ln b) \sum x^2 \quad (2)$$

And find a and b where x = time and $x = 1, 2, 3, \dots, n$.

where a = estimated Y intercept

$(b-1)100\%$ = estimated annual compound growth rate (in percent)

Certain Conditions Regarding the Construction of Trend Estimation Equations

1. The trend estimation equation must always be accompanied by the details of the equation. Specifically, you must clearly state the **Origin** and the **Units** of x and y
2. For **Time Series Data** collected over a specific duration, to facilitate calculation, the time value is defined as the **midpoint** of that period. For example:
 - 2.1 **Annual Data:** The midpoint is set as **July 1st** of each year.
 - 2.2 **Semi-Annual Data:** The midpoints are set as **April 1st** (for the first half) and **October 1st** (for the second half).
 - 2.3 **Quarterly Data:** The midpoints are set as **February 14th**, **May 15th**, **August 15th**, and **November 15th**.

Example: Based on the statistics of foreign tourists entering Thailand annually (in millions of people) from B.E. 2555 – 2559, as shown in the table displaying the annual number of foreign tourists.

Unit: Million people.

Year	Number of Tourists (y)
2555	5
2556	8
2557	16
2558	30
2559	40

- 1) Determine the Trend Forecasting Equation (T) Determine the Trend Forecasting Equation Linear Model, Quadratic Model, and Exponential Model Define T time and $X = 1, 2, 3, 4, 5$.

Solution:

Year	Y	X	X ²	X ³	X ⁴	X ⁵	(X ²)*Y	ln Y	X*lnY	log Y	X*log Y
2555	5	1	1	1	1	5	5	1.609438	1.609438	0.69897	0.69897
2556	8	2	4	8	16	16	32	2.079442	4.158883	0.90309	1.80618
2557	16	3	9	27	81	48	144	2.772589	8.317766	1.20412	3.61236
2558	30	4	16	64	256	120	480	3.401197	13.60479	1.477121	5.908485
2559	40	5	25	125	625	200	1000	3.688879	18.4444	1.60206	8.0103
Total	99	15	55	225	979	389	1661	13.55155	46.13527	5.885361	20.03629

$$\sum X = 15, \quad \sum X^2 = 55, \quad \sum X^3 = 225, \quad \sum X^4 = 979$$

$$\sum XY = 389, \quad \sum X^2Y = 1,661, \quad \sum \ln Y = 13.55155, \quad \sum X \ln Y = 46.13528$$

$$\sum Y = 99, \quad \sum \log Y = 5.88536, \quad \sum X \log Y = 20.03629$$

* Chart Builder.

GGRAPH

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/GRAPHSPEC SOURCE=INLINE.

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DATA: Year=col(source(s), name("Year"))

DATA: Y=col(source(s), name("Y"))

GUIDE: axis(dim(1), label("Year"))

GUIDE: axis(dim(2), label("Y"))

ELEMENT: line(position(Year*Y), missing.wings())

END GPL.

GGraph

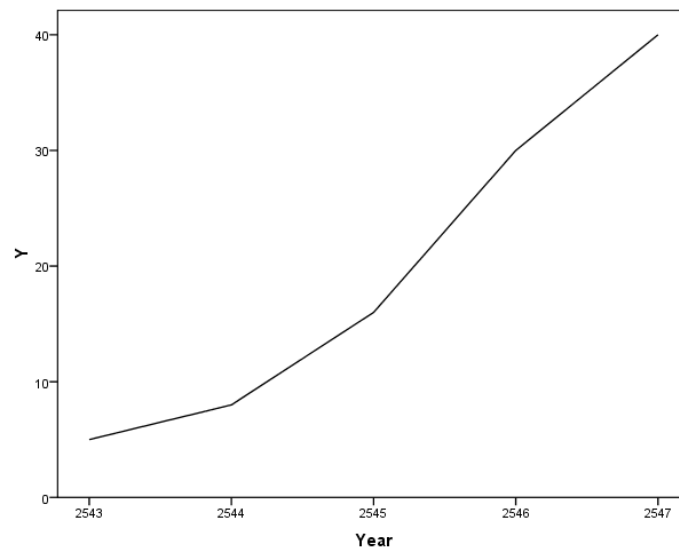


Figure 11: Graph showing the number of tourists from B.E. 2555 – 2559.

I. Linear Trend Equation

$$\hat{y}_T = a + bx \quad ; \quad X = 1, 2, 3, \dots, n$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{(5 \times 389) - (15 \times 99)}{(5 \times 55) - (15)^2} = \frac{460}{50} = 9.2$$

and

$$a = \bar{y} - b\bar{x}$$

$$= \frac{99}{5} - \left(9.2 \times \frac{15}{5}\right) = -7.8$$

$$\hat{y}_T = -7.8 + 9.2x$$

Where: $\hat{y}_T$ is the **estimated linear trend** of the annual number of tourists (in millions of people).

X is in units of **years** (1 unit of $X = 1$ year).

Origin : July 1, B.E. 2554 ($x = 0$).

II. Quadratic Trend Equation

$$\hat{y}_T = a + b_1x + b_2x^2$$

The Normal Equations obtained by the Method of Least Squares are as follows:

$$\sum y = na + b_1 \sum x + b_2 \sum x^2 \quad (1)$$

$$\sum xy = a \sum x + b_1 \sum x^2 + b_2 \sum x^3 \quad (2)$$

$$\sum x^2y = a \sum x^2 + b_1 \sum x^3 + b_2 \sum x^4 \quad (3)$$

Substitution into **Normal Equations**:

$$99 = 5a + 15b_1 + 55b_2 \quad (4)$$

$$389 = 15a + 55b_1 + 225b_2 \quad (5)$$

$$1,661 = 55a + 225b_1 + 979b_2 \quad (6)$$

Where X = time and $X = 1, 2, \dots, n$. Solving the system of equations using **Gauss Elimination** yields a , b_1 and b_2 as follows:

$$a = 2.2000$$

$$b_1 = 0.6286$$

$$b_2 = 1.4286$$

$$\hat{y}_T = 2.2000 + 0.6286X + 1.4286X^2$$

Where: $\hat{y}_T$ is the **estimated quadratic trend** of the number of tourists **annually** (Million people).

X is in units of **Years** (1 unit of X = 1 year).

Origin : July 1, B.E. 2554 ($x = 0$).

III. Exponential Trend Equation

$$\hat{Y}_T = ab^x$$

It can be transformed into a linear model by taking “ln” on both sides of the equation, yielding:

$$\ln y = \ln a + x(\ln b)$$

And find the values of a and b from the '**Normal Equations**' as follows:

$$\sum \ln y = n(\ln a) + (\ln b) \sum x \quad (1)$$

$$\sum x \ln y = (\ln a) \sum x + (\ln b) \sum x^2 \quad (2)$$

Substitution into **Normal Equations**:

$$13.55155 = 5(\ln a) + 15(\ln b) \quad (3)$$

$$46.13528 = 15(\ln a) + 55(\ln b) \quad (4)$$

Solve the normal equations to find a and b , where $X = \text{time}$ and $X = 1, 2, \dots, n$ Multiply Equation (3) by 3;

$$40.65465 = 15(\ln a) + 45(\ln b) \quad (5)$$

Take Equation (4) – Equation (5);

$$5.48063 = 10(\ln b)$$

$$\therefore \ln b = 0.548063 \Rightarrow b = e^{0.548063} = 1.7299$$

Substitute the value of $\ln b$ into Equation (3) to obtain:

$$13.55155 = 5(\ln a) + (15 \times 0.548063)$$

$$\therefore \ln a = 1.066121 \Rightarrow a = e^{1.066121} = 2.9041$$

$$\hat{Y}_T = 2.9041(1.7299)^x$$

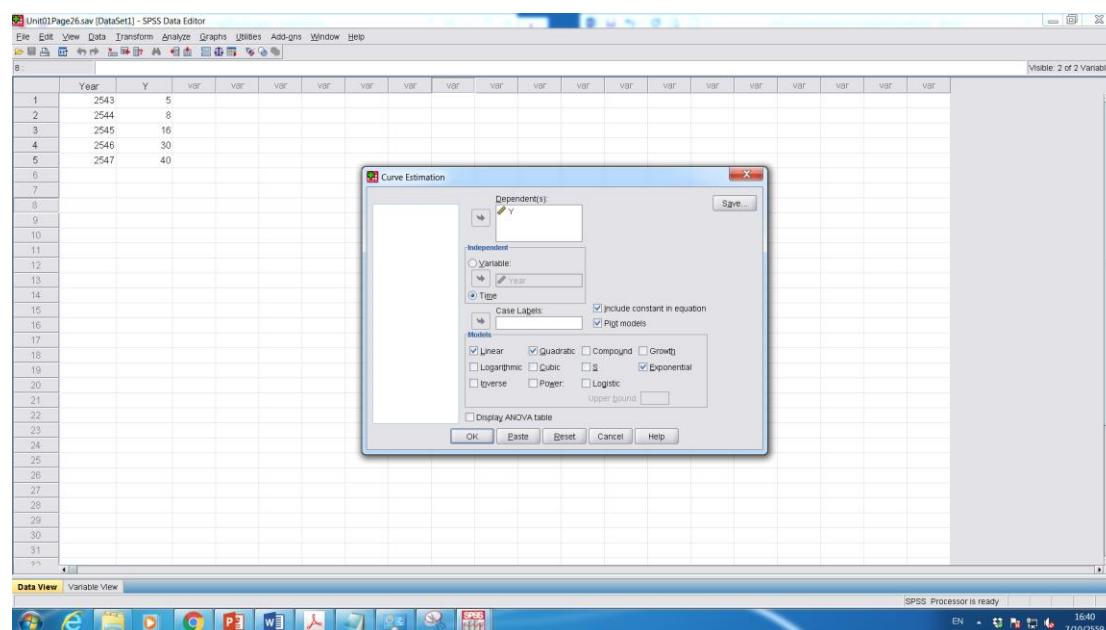
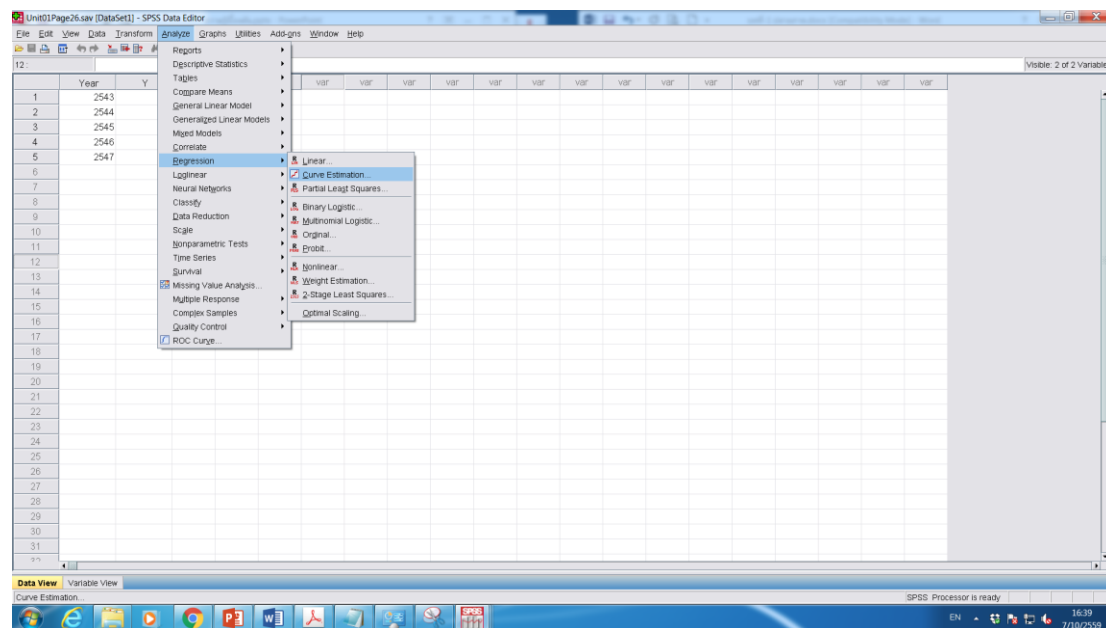
Where: $\hat{Y}_T$ is the **estimated exponential trend** of the annual number of tourists (**in millions of people**).

X is in units of **Years** (1 unit of $X = 1$ year).

Origin : July 1, B.E. 2554 ($x = 0$).

Chapter 4 Introduction to Quantitative Forecasting

Associate Professor Wikanda Phaphan, Ph.D.



* Curve Estimation.

TSET NEWVAR=NONE.

CURVEFIT

/VARIABLES=Y

/CONSTANT

/MODEL=LINEAR QUADRATIC EXPONENTIAL

/PLOT FIT.

Chapter 4 Introduction to Quantitative Forecasting

Associate Professor Wikanda Phaphan, Ph.D.

Curve Fit**Model Description**

Model Name		MOD_1
Dependent Variable	1	Y
Equation	1	Linear
	2	Quadratic
	3	Exponential ^a
Independent Variable		Case sequence
Constant		Included
Variable Whose Values Label Observations in Plots		Unspecified
Tolerance for Entering Terms in Equations		.0001

a. The model requires all non-missing values to be positive.

Case Processing Summary

	N
Total Cases	5
Excluded Cases ^a	0
Forecasted Cases	0
Newly Created Cases	0

a. Cases with a missing value in any variable are excluded from the analysis.

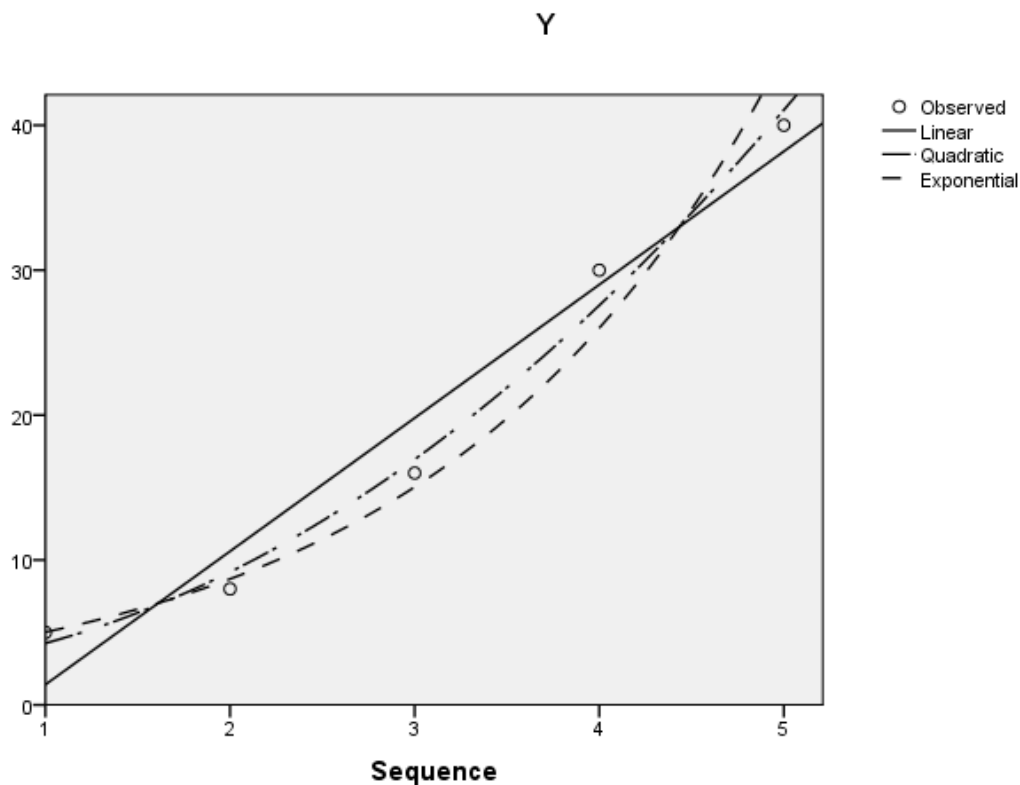
Variable Processing Summary

	Variables
	Dependent
	Y
Number of Positive Values	5
Number of Zeros	0
Number of Negative Values	0
Number of Missing Values User-Missing	0
System-Missing	0

Model Summary and Parameter Estimates

Dependent Variable: Y

Equation	Model Summary					Parameter Estimates		
	R Square	F	df1	df2	Sig.	Constant	b1	b2
Linear	.957	66.125	1	3	.004	-7.800	9.200	
Quadratic	.989	89.023	2	2	.011	2.200	.629	1.429
Exponential	.985	200.355	1	3	.001	2.904	$\ln b_1 = .548$	



From the results obtained, it is found that the **Quadratic Trend Estimation Equation**, which is: $\hat{y}_T = 2.2 + 0.6286x + 1.4286x^2$ best explains the number of foreign tourists. This is because it has the highest Coefficient of Determination (R^2), equal to 98.90%. Therefore, the Quadratic Model is selected as the annual trend estimation equation for the number of tourists in Thailand, where $X = \text{time}$ and $X = 1, 2, \dots$

Table 1.1 Comparison of the Number of Tourists (Y) and the Forecasted Trend Values ($\hat{Y}_T$) using Linear, Quadratic, and Exponential Models.

Year	Time (X)	Number of Tourists (Y)	Forecasted Trend Value ($\hat{Y}_T$)		
			$\hat{Y}_T = -7.8 + 9.2X$ (Linear)	$\hat{Y}_T = 2.2000 + 0.6286X$ $+ 1.4286X^2$ (Quadratic)	$\hat{Y}_T = 2.9041(1.7299)^X$ (Exponential)
2555	1	5	1.4000	4.2572	5.02398
2556	2	8	10.6000	9.1716	8.6913
2557	3	16	19.8000	16.9432	15.0356
2558	4	30	29.0000	27.5714	26.0109
2559	5	40	38.2000	41.0580	44.9979

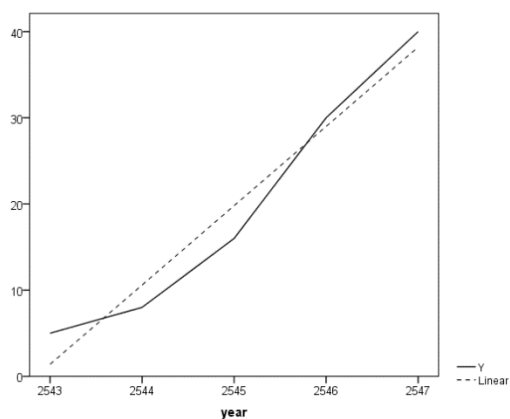


Figure A) Number of tourists (Y) and Linear Trend.

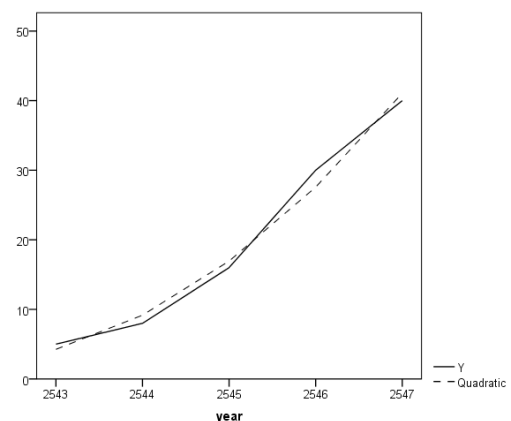


Figure B) Number of tourists (Y) and Quadratic Trend

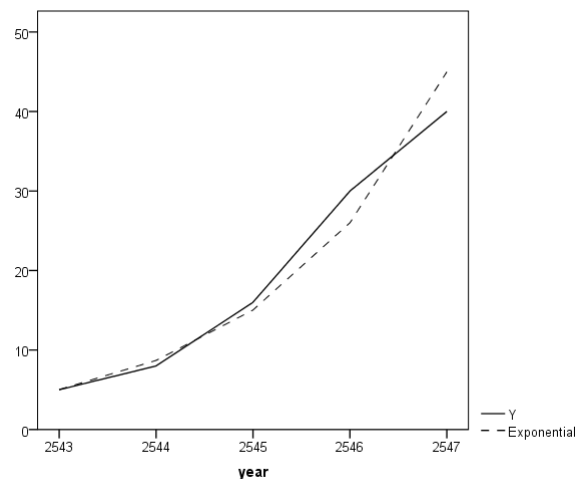


Figure C) Number of tourists (Y) and Exponential Trend.

2) If the analyst has calculated the 4 accuracy metrics based on the errors obtained for the trend estimation equations of all 3 models, resulting in the values shown in Table 1.2:

Table 1.2 This table shows the comparison of values MAD , MAD , MSE and MAD when using 3 trend estimation models.

Accuracy value	Trend Estimation Equation		
	1. Linear Model $\hat{Y}_T = -7.8 + 9.2X$	2. Quadratic Model $\hat{Y}_T = 2.2000 + 0.6286X + 1.4286X^2$	3. Exponential Model $\hat{Y}_T = 2.9041(1.7299)^x$
1. MAD	2.5600	1.2688*	2.1333
2. MSE	7.6800	1.9663*	8.4601
$RMSE = \sqrt{MSE}$	2.7713	1.4022*	2.9086
3. $MAPE$	27.2167%	9.2273%	8.1880%*
4. MPE	4.7167%	-0.0467%*	-0.4582%

Which equation would you choose as the trend estimation equation for the number of foreign tourists? Please provide your reasoning.

Solution

From the data, it can be seen that the values of MAD , MSE , $\sqrt{MSE}$ and MPE , which are used to measure forecasting error, are **lowest** for the **Quadratic Model**. However, when considering the **Exponential Model**, the forecast error measurement using the $MAPE$ value is the **lowest**. Therefore, the analysis may choose to use either the **Quadratic Model** or the **Exponential Model**, depending on which error measurement criteria is selected.

3) In which year was the number of foreign tourists most influenced by Cyclical and Irregular ($C \times I$) variations? Show the calculation method and explain the resulting values. (Use the trend estimation equation you selected from Question 2)

Solution

From the equation:

$$Y = T \times C \times I$$

It follows that:

$$C \times I = \frac{Y}{T}$$

Table 1.3 Comparison of the Number of Tourists (Y), Quadratic Trend Estimates(T), and the components of Cyclical and Irregular Variations ($C \times I$)

Year	Number of Tourists (Y : Million People)	Trend Estimate (T) Quadratic Model ($\hat{Y}_T$)	$(C \times I)(\%) = \frac{Y}{T} \times 100$ $(C \times I)(\%) = \left(\frac{T \times C \times I}{T} \right) \times 100$	Interpretation
2555	5	4.2572	117.45	Higher than normal by 17.45%
2556	8	9.1716	87.23	Lower than normal by 12.77%
2557	16	16.9432	94.43	Lower than normal by 5.57%
2558	30	27.5714	108.81	Higher than normal by 8.81%
2559	40	41.0580	97.42	Lower than normal by 2.58%

Therefore, the analysis reveals that the number of foreign tourists in B.E. 2555 was most influenced by cyclical and irregular variations, being 17.45% higher than normal.

4) Based on the trend estimation equation for the number of foreign tourists you selected in Question 2, forecast the trend for the number of foreign tourists traveling to Thailand in B.E. 2568 and B.E. 2569.

Solution

Year	Export Value (Y : Million Baht)	X
2555	5	1
2556	8	2
2557	16	3
2558	30	4
2559	40	5
2560		6
2561		7
2562		8
2563		9
2564		10
2565		11
2566		12
2567		13
2568		14
2569		15

From the **Quadratic Model**:

$$\hat{y}_T = 2.2000 + 0.6286X + 1.4286X^2$$

Where $\hat{y}_T$ is the estimated quadratic trend of the annual number of tourists (in millions of people)
 X is in units of **years** (1 unit of $X = 1$ year)

Origin : July 1, B.E. 2554 ($x = 0$).

For the year B.E. 2568, $x = 14$:

$$\hat{y}_T = 2.2000 + 0.6286(14) + 1.4286(14)^2 = 291.006$$

That is, the forecasted trend for the number of tourists in B.E. 2568 is 291.006 million people.

For the year B.E. 2569, $x = 15$:

$$\hat{y}_T = 2.2000 + 0.6286(15) + 1.4286(15)^2 = 333.064$$

That is, the forecasted trend for the number of tourists in B.E. 2569 is 333.064 million people.